

How the spaceplane lander works

From low Kerbin orbit to a stop on the KSC runway: one deorbit burn and then 1,500 km of unpowered flight, from a kRPC script

The vehicle starts in orbit with an engine, a little propellant and no further thrust after one burn. The autopilot chooses when and how hard to burn, flies a hypersonic entry with angle of attack and bank, spirals off its leftover energy in a circle over the field, glides down final, flares and brakes to a stop. The runway is 2.4 km long and 70 m wide. On the target vehicle, a Mk3 shuttle, it lands intact on the runway in about three flights of four from a save taken over the field. From orbit it is still the open problem.

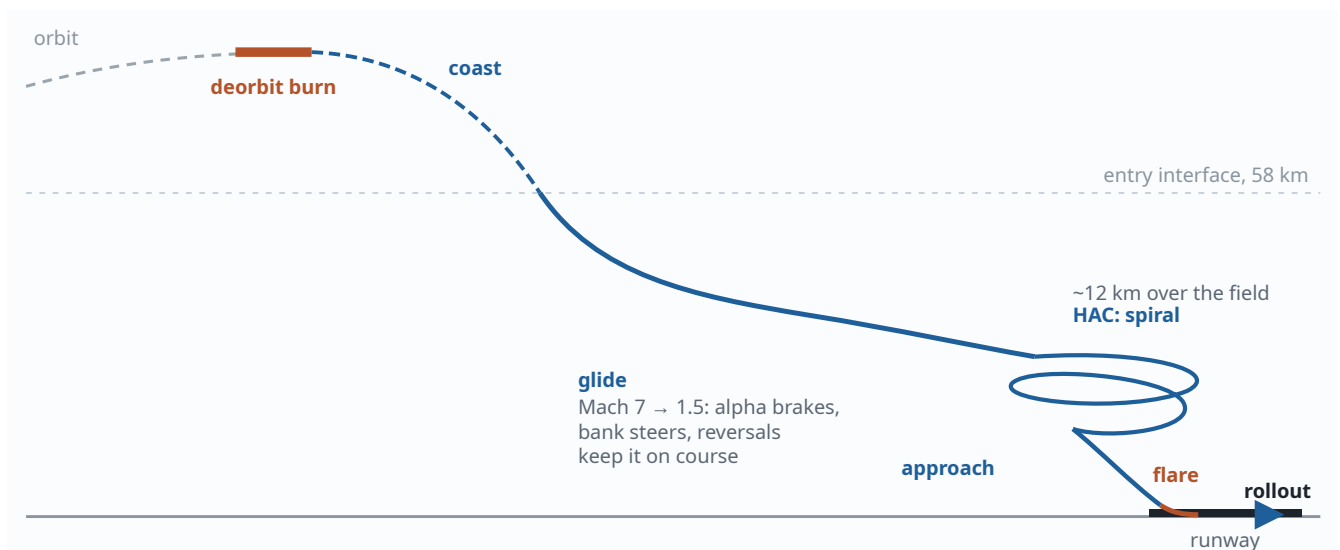


Figure 1. The flight profile, heavily not to scale: the glide covers well over a thousand kilometres, the circle over the field a few tens. Only the deorbit burn uses the engine.

The architecture

The spaceplane shares the booster lander's skeleton. One phase machine (`autopilot.py`) is the only code that talks to the game. **Every tick it re-propagates the vehicle to the ground** with a local RK4 integrator (`trajectory.py`) that models gravity, lift, drag and the planet's rotation, and every phase decision comes from that fresh prediction rather than from a stored plan. The control laws are pure functions in `guidance.py`. `environment.py` holds the air, the runway and the airframe's aerodynamic table, so the integrator never makes a remote call.

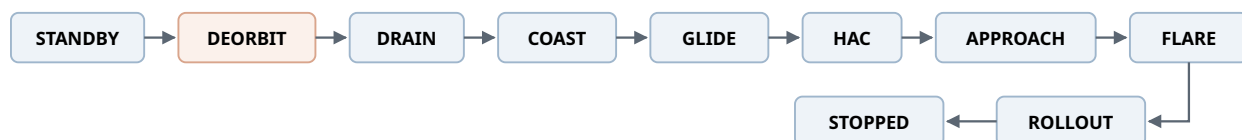


Figure 2. The phase machine. With the default `DRAIN_BEFORE_BURN`, DRAIN runs just before the burn: surplus propellant is dumped in orbit, leaving a reserve the burn spends. Phases hand over at altitudes and geometry, never at a miss distance.

Three ideas recur through the whole design.

- **Measure the airframe; don't type it in.** The lift and drag table is swept in STANDBY with kRPC's `simulate_aerodynamic_force_at`, 182 probe calls in under a second, so the first prediction already uses this vehicle's coefficients. The probe takes the airflow as an argument, so each cell is measured at exactly the angle of attack it tabulates. The vehicle's own axes are checked against `Vessel.direction` before anything is trusted. The highest angle of attack the airframe can actually *hold* is learned in flight (see the glide).
- **Surplus energy is recoverable; a deficit is not.** A glider can always throw energy away by banking, pitching up or S-turning, but it can never make more. So every aim in the flight sits deliberately on the long side, and the gates are placed above the best glide from them, not below.
- **The predictor flies the law the vehicle flies.** The speed cap, the bank-reversal behaviour and the alpha ceiling are applied inside the propagator as well as in the control loop. A prediction of a law nobody flies is a prediction of a trajectory nobody flies.

Deorbit: choosing the one burn

In orbit the autopilot asks each tick whether the runway is within deorbit range, time-warping while it isn't. Inside the range it searches candidate burn sizes, propagating a full entry for each, and commits when one works. What “works” means is the important part.

The glide that follows has a finite **window** of reachable arrivals. Its shortest is the steepest, highest-drag entry the glide solver is allowed to fly, and its longest is the most lifting one. `guidance.deorbit_window` propagates those two corners, and the burn is chosen to put the target **in the middle of the window**. Because the corners are exactly the commands the glide may issue, the window is the glide's real authority rather than a guess about the airframe. This replaced a fitted “aim long by X% of the arc” constant that had to be re-fitted five times in one session, each time forced by an unrelated change elsewhere.

The burn itself is closed-loop: it points retrograde, and instead of asking “how much more?” it asks each tick whether the trajectory *already* does the job (one propagation, not sixteen). The engine's thrust limiter is turned down so small burns last at least three seconds and can be cut off accurately. The flip to burn attitude uses RCS, because it is on the critical path; flying it on reaction wheels alone cost 50 km on every arrival.

WHY THE BURN MATTERS SO MUCH

One metre per second at the entry interface is worth roughly 11 km on the ground. Fifteen flights from three saves taken *at* the interface repeated to within 2.3 km, against 14.8 km for whole flights from orbit: as variance, **97.6% of the along-track scatter is made before the entry starts**. The single largest fix to it was not in guidance at all. The propellant drain valve had been firing *after* the burn's closed loop stopped looking, giving a small, unmeasured push. Draining before the burn closed the gap between entry states.

Coast

After the burn the vehicle falls to the 58 km entry interface. It is already in the entry attitude (22° angle of attack), because an attitude set up in vacuum costs nothing to hold, while one set up in thickening air has to be fought for. RCS is permitted, and time warp is used where nothing needs control.

Glide: angle of attack brakes, bank steers

From the interface down to about Mach 1.5, the glide re-solves its two controls every tick to arrive over the field. The work is split between them because the measurements force it:

CONTROL	WHAT IT DOES
angle of attack	Downrange and energy. Drag at 30° is ten times drag at 0°. Solved against the predicted along-track miss.
bank magnitude	Rolling lift off the vertical lets the vehicle sink into thicker air, which is what really slows it. Takes whatever the angle of attack leaves.
bank sign	The <i>only</i> cross-track authority in the whole flight. The sign flips (a bank reversal) whenever the heading error to the runway leaves a deadband, and the deadband narrows as the range shrinks.

The obvious design, solving both controls together through a 2×2 Jacobian as the booster's wing does, failed here. The sensitivity of range to angle of attack **changes sign during the entry**: at 80 km more alpha lengthens the flight, and by 45 km it shortens it. Range is also non-monotone in both controls at the shallow end: maximum L/D gives the *shortest* flight, and a little bank lengthens it. The solver brackets to stay on the monotone side of each.

The propagator models a reversing entry as its average: vertical lift reduced by $\cos(\text{bank})$, with no net sideways lift. Modelling one fixed lean instead predicted a huge curving arc whose cross-track sign flipped every cycle, which commanded a vehicle rolling through 40° twice a second.

The ceiling the airframe will actually hold

In dense air the vehicle doesn't hold what it is told: asked for 20° at Mach 4 it delivers 15–18°, and the shortfall grows with dynamic pressure. That is a **saturation**, $\text{achieved} = \min(\text{command}, \text{holdable}(q))$, not a gain, and the difference matters. With a proportional model the solver simply asks for more, the factor cancels, and the prediction believes a lift that never arrives. kRPC won't report a pitching moment, so the ceiling can't be probed. `trajectory.Holdable` learns it instead from every tick's command-versus-achieved pair, binned by dynamic pressure. A bin with no evidence returns `None`, and queries beyond anything seen extrapolate *downward* only. Predicting too little lift makes the burn larger, and surplus is the recoverable side.

What it misses (2026-10-06). A ceiling learned only by saturating is unknown in air the vehicle has not reached. A glide that tracks 40–44° early is planned at that alpha into 3–5 kPa, where the shuttle holds 23–28°. Its measured drag there is 0.59–0.80 of the predicted drag (0.83–1.07 on flights that arrive on profile; the table at the *achieved* alpha is within 4% on all of them), and it reaches the cone 4–14 km long. `HOLDABLE_PRIOR` (off) seeds the bins from the craft's own logs (`tools/holdprior.py`). It removed most of the long arrivals, but handed the cone over low on every shuttle orbit and took the old craft 6–8 km short. A median over dynamic pressure alone is not the plant.

Why it misses: the attitude controller, not the airframe (2026-10-07). Holding 36° at 2–4 kPa takes the shuttle's whole pitch input, and kRPC holds most of it in an integrator. kRPC 0.6 keeps its pitch and yaw integrators in a frame that does not roll with the vehicle, so at the bank reversal near 38 km a wound-up pitch integral is swung partly onto yaw. On the long flights the pitch input falls from 1.0 to 0.4 and stays there for 30 s against an 8–12° pitch error, with 8–10° of sideslip. `GLIDE_PITCH_OFFLOAD` (off) carries the standing pitch input as a manual trim in the body frame, which no roll can rotate, the way an aircraft autopilot offloads sustained elevator to the trim tab. Two rules came out of flying it. It never learns from a saturated read-back (chasing one pinned the elevons nose-up), and it only works above Mach 3 (held lower, the glide arrived 2 km low). On the bimodal save it took the cone arrival

from +4.4 km sd 3.7 to +1.1 sd 0.8, and survival from 46/56 to 55/56. Landings stayed level, because the energy chain below now decides them: the glide aims a 12 km crossing (12.9 km of energy height) where the cone needs about 18.5, and inside the cone some flights' speed collapses to 70–90 m/s.

Arriving at the right speed

Hold the entry angle of attack to the end and the vehicle slows to that angle's equilibrium speed, which is below stall. A speed cap therefore takes over near the end and uses angle of attack to hold the approach speed. It is two-sided below a set Mach number, because a one-sided cap disengages exactly when the vehicle dives, and a dive turns surplus height into speed that nothing after the gate can spend. One flight arrived at the approach doing 219 m/s, 74° off the runway heading.

HAC: stop asking the entry for an exact energy

This is the change that took the landing from tens of kilometres to hundreds of metres. It works by changing what the entry is asked to do, not by doing it more accurately. A straight-in arrival has to be at one point, one height and one energy after 1,500 km of unpowered flight, and the burn alone scatters that by ~15 km. The Space Shuttle didn't try. It arrived *over* the field with height in hand and spent the surplus turning round a **heading alignment cone** (HAC), a circle tangent to the extended centreline. Rolling out of the circle is lining up, so alignment and energy management become the same manoeuvre.

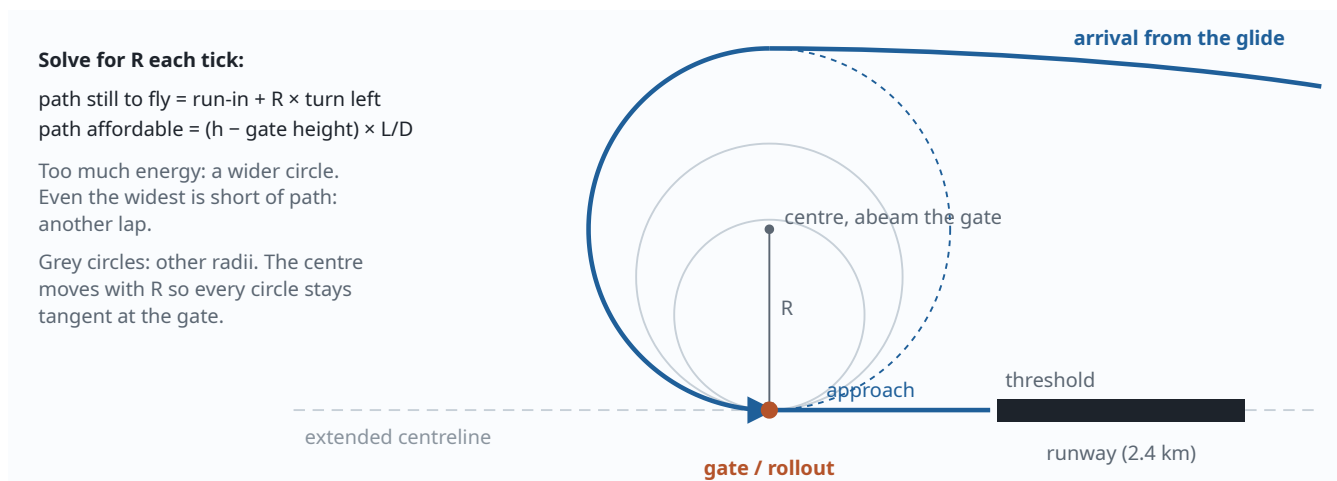


Figure 3. Plan view of the heading alignment cone. An energy error becomes a radius, and a radius is something the vehicle can just fly. The circle is entered only below Mach 1.5: turn radius goes as v^2 , about 9 km at Mach 0.9 and 250 km at Mach 4.7.

Making this work cost several flights, and each lesson is now a rule in the code:

- **The centre moves with the radius.** With the centre pinned 8 km abeam, every other radius missed the gate sideways by (8 km – R). Three flights landed 0.2–1.6 km off the centreline of a 70 m runway. With the centre moved, the next four landed within 200 m.
- **The tangent point is reached travelling the right way round.** Both tangent points satisfy the arithmetic. With the wrong sign, a vehicle already lined up 30 km out was costed a 149° turn and abandoned the cone.
- **Lined up is not ready.** Reaching the rollout with a lap still owed, or with more height than the approach can lose, is not an exit. Two early flights handed the approach a vehicle 12 km up and floated 10 km past the field.
- **The entry's target moves up.** The glide now aims at a point ~12 km over the gate instead of at the gate itself. The vehicle crosses the field at Mach 2.9 at 19 km, so the cone has to be delivered where

it can be flown, and the raised target gives back ~19 km of required range the entry was carrying as surplus anyway.

Trim drag is the speedbrake (2026-10-07)

Every shuttle cone flies with the elevons near their nose-up stop: a standing pitch input of +0.55–0.76, at L/D 1.3–1.6. After the drain, all the propellant the shuttle keeps (~2 t) sits in its nose tank, 13–15 m ahead of four empty ones, and nothing ever commanded moving it. `PROPELLANT_TRIM` pumps it aft through kRPC's resource transfer (~2 s for the whole load) whenever the standing input is nose-up. It works as a trim: the input falls to +0.15, the bank tracks its command, and the cone's glide ratio rises from 2.0 to 2.6. As a landing aid it fails. The trim drag was the only speedbrake this craft has (its measured spoiler dumps lift instead of adding drag), so the trimmed vehicle arrives 7–15 km long, and pumping during the high-alpha glide pitches it up into a deep stall. As an *energy* control (aft only when the cone is short of height) it centres the arrivals on the runway for the first time, but still loses a third of its vehicles. It is off by default.

Who holds the trim (2026-10-07 evening)

The cone's remaining short misses were not a planning error. Late in the spiral the vehicle held 6–11° of alpha against a 0–2° command, with a 7–9° pointing error and an *unsaturated* pitch input, for up to a minute and a half. kRPC's own diagnostic log explains it. Its attitude controller keeps its integrators in a "roll-invariant" frame that it carries along with the nose, and a spiral twists that frame against the body (34° on a vehicle flying wings level). The nose-up trim stored there then comes out as pitch *and* yaw error and unwinds over minutes. Holding the trim elsewhere in software failed twice. What worked was moving it into hardware. `CANARD_TRIM` takes the two nose canards off kRPC (kRPC's per-surface `deflection_override`) and moves them itself as a trim that drives the standing pitch input to zero. The canards are also the efficient place for it. Trimming with every surface at full nose-up input costs a third of the lift and more than doubles the drag, while canard trim *adds* lift. Two limits were measured first: the canards stall (their nose-up effect is gone by ~24° of alpha), and the law only engages settled, under 15°. With the cone planning from its own measured glide ratio, the bimodal orbit landed **31 of 48** on the runway, against about a third. It is off by default because the inclined and high orbits land long with it so far.

What the glide hands the cone, and why the cone mis-prices it (2026-10-08)

The cone flies about 27 km of path from every entry, whatever radius it picks, and its only real energy control is how long it stays at its alpha cap. So it can absorb at most ~20–21 km of entry energy height ($h + v^2/2g$). Entries above that land 4–6 km long. The glide over-delivers because below Mach 3 it commands ~40° of alpha and the shuttle holds only 57–75% of it, so its propagator predicts more drag than happens. A Mach-banded `HOLDABLE_PRIOR` (the alpha this craft held, per Mach band and dynamic pressure, from its own logs) fixes the prediction and cuts the mean entry energy from 18.9 to 17.4 km. Landings did not move (21 of 36 either way): the stop tracks the cone's rollout surplus, not its entry energy.

The cone mis-prices for two reasons. Its aero table is re-probed in flight with the control surfaces at their *present* deflection. At cone entry the subsonic rows describe an airframe trimmed for Mach 3, which prices the descent at L/D 0.85–1.72 where the cone will fly 3.3–3.7. And the wing as flown, trimmed, makes its peak lift of ~139 m² near 12°, about two-thirds of what the table claims. The cone spends most of its upper half at 15–17°, past that peak, at L/D ~1.2, braking toward a fixed *true*-airspeed target. Pricing the plan off the flown polar (`HAC_LD_FLOWN_POLAR`, off) works as intended: the cone sees its surplus from entry. That exposes the next limit. The surplus falls between the widest

circle without a lap and a minimum lap (~12.6 km of path), a gap no radius closes. Holding the cone at constant indicated airspeed (`HAC_SPEED_EAS`) was refuted on the farm: 1 of 12.

Approach: hold speed, not path

A glider can't choose its speed and its flight path independently: pick an angle of attack and the polar fixes both. The one to choose is **speed**, because arriving slow can't be recovered. The first approach law used the throttle-aircraft logic of commanding alpha from sink-rate error, which is backwards here. Arriving high, it unloaded the wing and traded the height for speed, and offline reached the threshold at 102 m/s and 36 m/s of sink.

The current law commands the *descent angle* that holds the target speed, from $dv/dt = g \sin\theta - D/m$. That lets it make speed by steepening as well as lose it. Surplus height is spent with S-turns that command a track angle, $\arccos(\text{straight path} / \text{path the height can pay for})$, rather than a bank. The target speed is deliberately **fast**, $2.25\times$ stall. The load factor available to arrest the sink in the flare grows as the *square* of speed ($n_{\max} = (V/V_s)^2$), while the sink to be arrested grows only linearly, so a faster approach makes the flare strictly easier. Horizontal speed is the cheap axis, and the runway is where it is meant to go.

Sharp turns as a brake (2026-10-07 late, off)

The user suggested a split-S to bleed altitude. A literal split-S turns height into *speed*: inverted, lift and gravity both pull down. It also points the vehicle away from the runway, and the turn back costs more height than the surplus. The useful half of the idea is that **a hard turn is drag**. This airframe's drag rises fivefold from 2° to 16° of alpha, and banking puts the extra lift sideways instead of ballooning. `APPROACH_SHARP_TURN` chooses the alpha whose drag spends the remaining surplus before the S-turn stops. It keeps the vertical lift a steady descent at the present speed needs and banks off the rest. Two other references for that vertical lift were flown and failed: the speed law's own lift (turning instead of drag, one flight landed backwards) and one g (it spent speed, not height). `HAC_SPIRAL_DUMP` flies tight descending 360s over the gate when the cone lines up kilometres high. In one flight it took a save that otherwise stops 14 km long to 62 m from the runway's centre. But a lap costs ~3.9 km on this airframe at any speed, so smaller surpluses can't use it, and twice it became a spiral dive. Neither is ready.

`HAC_GATE_STRETCH` (off) tried to spend a surplus smaller than a lap continuously: while the plan has height to spare, the rollout point moves out along the extended centreline and rises by the approach's own glide, so the extra height becomes a longer final. It was flown 12 against 12 on the cone saves and changed nothing. The high saves enter the cone straight in, on the centreline outside the gate, and from there moving the gate adds no path. It only reprices path the vehicle flies anyway. Only a turn away from the runway adds path there, and the cheapest such turn is a lap.

Flare and rollout: no law may step

Four separate laws in the last few hundred metres were commanding *steps*, and each one destroyed the vehicle on a runway it had already reached. The common rule: **ramp every command, and near the ground reference it to the runway, not the airflow**.

- **Flare (from 50 m).** The flare tracks a sink rate of $\sqrt{(v_{td})^2 + 2ah}$, easing smoothly to the touchdown rate. Its arrest demand divides by height, so a floor stops it from demanding maximum lift over the last two ticks, and a touchdown attitude caps pitch short of a tail strike. Below 30 m it aligns the nose with the runway, not with the velocity. Pointing at the velocity landed the vehicle crabbed: 23° of sideslip at the first rollout tick, 51° two seconds later, gear torn off.

- **Centreline capture.** The approach asks for the lateral closure that brings cross-track to zero *when the wheels arrive*, not just “inside the offset left”. The flare has only about 1.5 s of lateral authority, so it inherits whatever drift it is handed.
- **Rollout.** Ramp the nose down from the attitude it touched down in, instead of commanding zero alpha at 90 m/s (a nose-over) or nothing at all (a tail scrape). Full main-gear braking from touchdown, nosewheel steering onto the centreline on cross-track *plus* its rate times a 3 s lead (cross-track alone is an undamped double integrator: every shuttle rollout weaved $\pm 25\text{--}80$ m and some left the tarmac; with the lead they close monotonically and stop within 7 m), ground spoilers. STOPPED hands the vehicle back.

Where it stands (2026-10-08)

ndash;133nbsp;m/s with 60ndash;90nbsp;m/s of sink, from handovers ~ 1 km high. Capping the sink (`APPROACH_SINK_GUARD`) stopped the dives but stalled the vehicles short instead: the energy has to be managed, not the sink.

SEGMENT	STATE
deorbit + entry	Works on both airframes. The shuttle's arrival error went from +111 km to +3 km after attitude and burn fixes, including the engine thrust limiter, which had never been commanded.
landing, from saves over the field	Shuttle on the runway intact in 23 of 30 flights across five save states (36 with a sixth, hard one included: 23/36). Touchdown cross-track is now a median of 13–20 m.
landing, from orbit	19 of 36 orbital flights now land on the runway intact, across three entry states (from 3 of 36). Two fixes in the cone did it. The cone's entry aim is derived from the shuttle's own glide, and its weave no longer opens with a 100°+ bank reversal at Mach 0.9 and 42° of alpha, which had been tumbling the vehicle. The cone now hands over within ±500 m of height in 23 of 36 flights. The touchdown aim moved from the far threshold to 1800 m, but that number was fitted on the shuttle (the old craft overruns on it) and is to be replaced with one derived per vehicle. Blocker: on one orbit the glide is bimodal, with 7 of 12 arriving 3–10 km long after running out of bank authority near Mach 4.
glide pitch offload (off)	Carrying the glide's standing pitch input as a body-frame trim removes the long mode caused by kRPC's roll-invariant pitch integrator. On the bimodal orbit it keeps 50 of 52 vehicles intact and lands 23 of 52 on the runway. Every miss is short: the cone runs out of height. The propellant pump has not yet beaten it.
canard trim (off)	On the bimodal orbit, with the cone's measured glide ratio: 31 of 48 on the runway (same-day baseline about a third), the short cluster gone. On other orbits so far: inclined 1/6, high 2/6, both landing long; an eccentric orbit arrives 41 km long at the cone, an entry problem. Next: runtime crossrange, and spending surplus height with alpha rather than a dive.
sharp turns / spiral (off)	The cone saves, with the canard trim, roll out lined up 4–6.8 km high and land 3–14 km long. The cone hands over anything short of a whole lap (~8 km of height), and the approach can spend only ~0.8. The tight-spiral dump fixed one of three such flights and dived on the other two. Moving the rollout out (HAC_GATE_STRETCH) was null: the high arrivals come in straight on the centreline. Next: drag at the cone's speed, or an entry aim that puts high arrivals onto the circle.
cone energy (2026-10-08, off)	Mach-banded alpha prior: handover energy down 1.5 km, landings unchanged (21/36 v 21/36 over three orbits). Flown-polar pricing: the cone sees its surplus and hits the no-lap/lap gap, 3/12. Constant-IAS cone refuted, 1/12. Next: close the gap by planning the lap at entry, or aim entries inside the no-lap band.
split rudder (2026-10-08, off)	The twin fins split as a speedbrake: approach drag doubles with no yaw or roll. As the cone brake it cut long landings 5 to 1 in its first batch (7/12 v 5/12). On the runway it adds to the elevon/canard spoiler.
approach dives (open)	About one flight in five enters the flare at 95
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crossrange (measured)	Holding 45° of bank from entry to Mach 1 moves the vehicle 57–72 km sideways; 60° moves it 92–95 km, plus whatever the subsonic glide adds. The lost downrange is bought back by burning later.
code	139 experiment flags retired and about 11,000 lines removed, with the guidance laws verified unchanged. The stall speed is now always the vehicle's own, from its swept table.
touchdown survival	The shuttle's wings were breaking off on contact. The cause was the <code>rigidAttachment</code> flag on the two delta-wing parts: with it off, 22 of 24 touchdowns kept both wings, against ~8 of 24.

How it is measured

Changes are flown on a farm of six headless KSP instances, with the autopilot unmodified. A batch is n flights of one configuration, not one flight of n configurations: one unchanged configuration scatters 10–15 km along-track from orbit, and an 8-flight batch resolves a factor of two and nothing finer. Arms are interleaved round by round so drift in the farm cancels. Every log records the settings that differ from the defaults *and* a fingerprint of the defaults, and ends with a `loop rate` line showing whether each phase's control loop actually ran at the interval it asked for. Landing work is flown from saves taken over the field, since a landing can't be measured through 15 km of entry scatter. A headless simulator (`kspSim`) screens ideas about ten times faster, and the farm confirms them.

Written 2026-10-05, updated 2026-10-06, from `docs/spaceplane/design.md`, `failures.md`, `journal.md`, `HANDOFF.md` and the code at the end of 2026-10-06 (defaults fingerprint `d91cd0ad`). Figures are from the logs cited in those documents.